

Physics Baselines, Production MLP, and SVGP

2026-05-21 full-clean / uncensored CDF / P50–Q1 evaluation

Linan Jiang

Aalto University

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Why This Deck Was Refreshed

Old version

The earlier baseline deck described an early-May intermediate model and a narrative centered on improvement over an early GP baseline.

Current thesis protocol

The updated comparison uses the 2026-05-21 evaluation artifacts: production MLP five seeds, the mu-anchor/raw-uncertain variant, Stage-3 SVGP, and refreshed Hiroyasu–Arai / Naber–Siebers curve baselines.

One-line result

The production MLP strongly beats HA/NS physics baselines. SVGP remains slightly better on point RMSE/MAE. The MLP is the more conservative uncertainty surrogate for screening.

Evaluation Contract

Protocol	Points	Conditions / trajectories	Role
Full-clean CDF	692 942	24 316 trajectories	Main headline comparison on the common clean CDF set.
Conservative uncensored CDF	542 565	592 conditions	Removes conservatively flagged FOV right-censoring and signal-decay time bins.
P50 observed	4213	495 conditions	Condition-level P50 observed target for robust mean-head checks.
Q1 extrapolated grid	21 527	495 conditions	P50/Q1 fit grid for extrapolated lower-quantile behavior.

The protocols answer different questions: full-clean CDF measures overall point accuracy, uncensored CDF controls censoring bias, and P50/Q1 checks condition-level robustness and extrapolation.

Source: [Thesis/generated/baseline_comparison_20260521/summary.json](#)

Models Compared

Model	Setup
Hiroyasu–Arai calibrated	Empirical curve baseline recalibrated under the current clean protocol.
Naber–Siebers delay Production MLP	Empirical curve baseline with delay handling, also refit. Stage-3 anchor_off, seeds 7/17/42/99/2024, reported as mean $\pm$ std.
mu-anchor raw-uncertain prodKD	Same data and teacher warm start, but with the mu-anchor/raw-uncertain/product-KD variant.
Stage-3 SVGP	Sparse heteroscedastic GP baseline, seed 42, evaluated on the same tables.

Boundary

This is a same-data evaluation comparison, not a LONO comparison. LONO remains a separate generalization stress test in the stage-1/2/3 ablation decks.

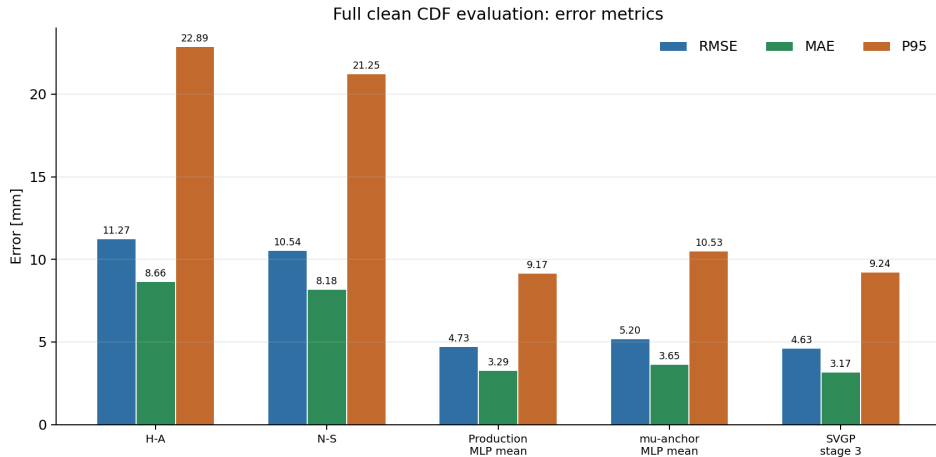
Full-Clean Headline

Model	RMSE	MAE	Bias	P95	Cov1
Hiroyasu-Arai	11.268	8.658	+0.288	22.891	0.615
Naber-Siebers	10.545	8.177	-0.442	21.253	0.637
Production MLP mean(5)	4.735	3.293	+0.626	9.167	0.867
mu-anchor raw-uncertain	5.200	3.654	-0.585	10.525	0.834
SVGP Stage-3	4.628	3.173	-0.070	9.242	0.703

The production MLP more than halves HA/NS RMSE, but SVGP is still the point-accuracy winner. MLP coverage is substantially more conservative.

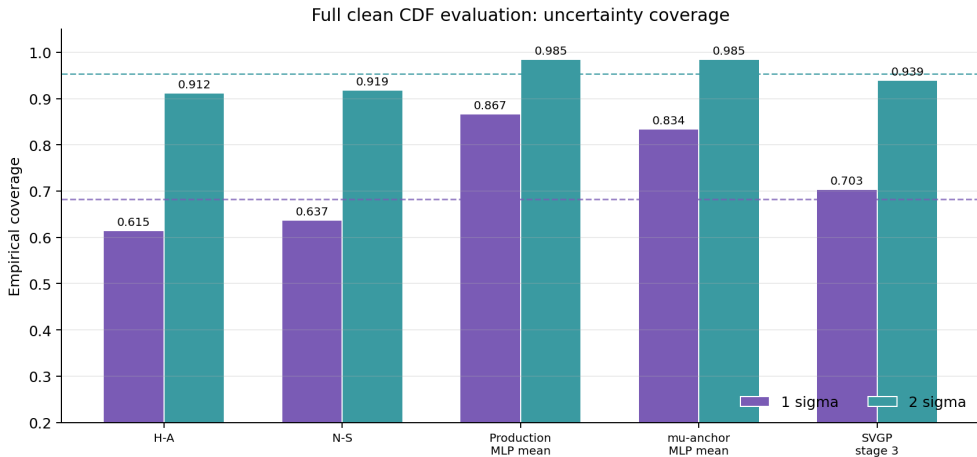
Source: Thesis/generated/baseline_comparison_20260521/headline_full_clean.csv

Full-Clean: Error Metrics



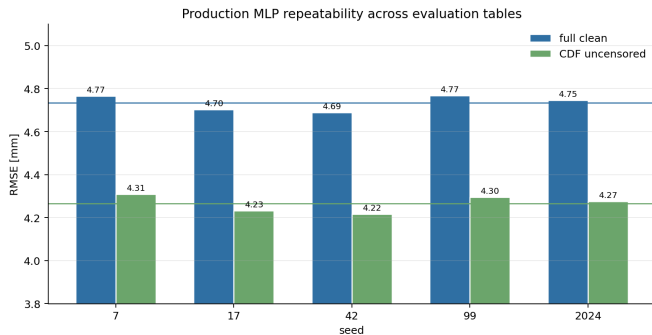
Source: Thesis/images/baseline_comparison_20260521/full_clean_metric_bars_rmse_mae_p95.png

Full-Clean: Coverage



Source: Thesis/images/baseline_comparison_20260521/full_clean_coverage_comparison.png

Production MLP Five-Seed Stability



Full-clean aggregate

- $\text{RMSE} = 4.735 \text{ mm} \pm 0.037 \text{ mm}$
- $\text{MAE} = 3.293 \text{ mm} \pm 0.039 \text{ mm}$
- $\text{Bias} = +0.626 \text{ mm} \pm 0.112 \text{ mm}$
- $\text{Cov1/Cov2} = 0.867/0.985$

Reading

This is not seed luck: the five-seed RMSE spread is only about 0.04 mm.

Source: Thesis/generated/baseline_comparison_20260521/production_mlp_full_clean_per_seed.csv

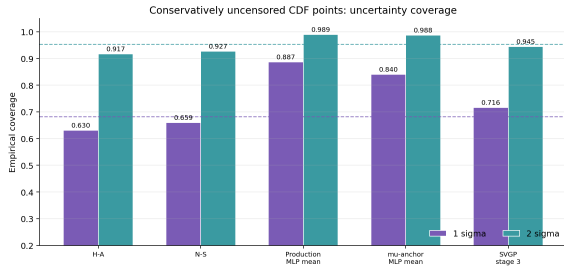
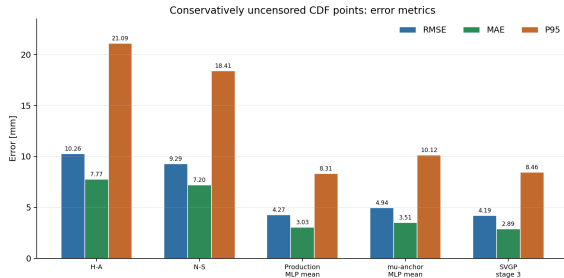
Conservative Uncensored CDF

Model	RMSE	MAE	Bias	P95	Cov1
Hiroyasu-Arai	10.261	7.767	+2.152	21.095	0.630
Naber-Siebers	9.286	7.195	+0.628	18.408	0.659
Production MLP mean(5)	4.265	3.032	+0.562	8.307	0.887
mu-anchor raw-uncertain	4.941	3.505	-0.868	10.123	0.840
SVGP Stage-3	4.193	2.894	-0.040	8.461	0.716

After conservative censoring control, production MLP and SVGP remain close: SVGP has slightly lower RMSE, while MLP keeps wider coverage.

Source: Thesis/generated/baseline_comparison_20260521/fixed_table_group_summary.csv

Uncensored CDF: Error and Coverage



Source: Thesis/images/baseline_comparison_20260521/cdf_uncensored_*.png

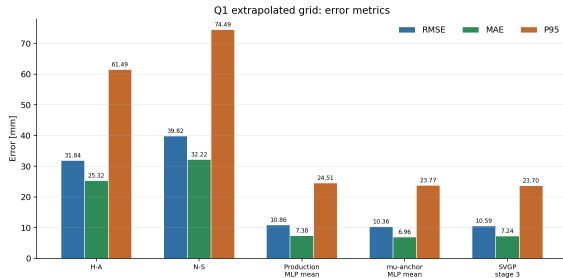
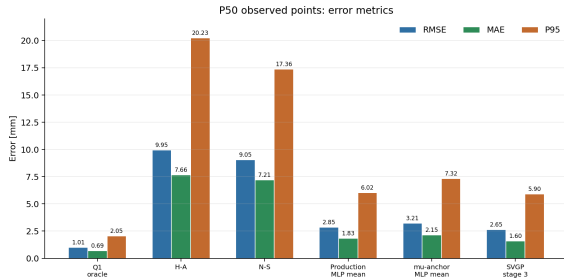
P50 and Q1 Protocols

Model	P50 RMSE	P50 MAE	Q1 extrap. RMSE	Q1 extrap. MAE
Q1 oracle	1.010	0.689	–	–
Hiroyasu–Arai	9.948	7.662	31.837	25.316
Naber–Siebers	9.048	7.206	39.818	32.216
Production MLP mean(5)	2.848	1.826	10.864	7.380
mu-anchor raw-uncertain	3.212	2.147	10.358	6.965
SVGP Stage-3	2.649	1.603	10.591	7.236

SVGP leads on P50 observed. The mu-anchor variant helps slightly on Q1 extrapolation, which is useful as a secondary result rather than the main headline.

Source: [Thesis/generated/baseline_comparison_20260521/fixed_table_group_summary.csv](#)

P50 Observed and Q1 Extrapolated



Source: Thesis/images/baseline_comparison_20260521/p50_observed_*.png;q1_extrapolated_*.png

- 1 Do not claim that the MLP is ahead of SVGP; the latest results do not support that.
- 2 Do claim that production MLP is far stronger than HA/NS on all clean, uncensored, and P50 protocols.
- 3 Do report SVGP as the fair ML baseline and point-accuracy leader.
- 4 Do emphasize that MLP uncertainty is more conservative and useful for downstream screening.
- 5 Keep mu-anchor/raw-uncertain as a Q1 extrapolation supplement, not the main headline.

Item	Path
Slides source	Thesis/slides/baseline_comparison_slides_en/baseline_comparison_slides_en.tex
Figures	Thesis/images/baseline_comparison_20260521/
Generated tables	Thesis/generated/baseline_comparison_20260521/
Fixed-table report	MLP/baseline/comparison_reports/stage3_fixed_table_eval_with_hans_20260521/
Full-clean MLP vs SVGP	MLP/baseline/comparison_reports/mlp_production_stage3_vs_svgp_stage3_full_clean_20260521/
Figure script	Thesis/generated/baseline_comparison_20260521/make_figures.py